two will preserve an independent reality." There is just one, unique, absolute, four-dimensional spacetime in our Universe; and a warpage of everyone's time and everyone's space must show up as *a warpage of Minkowski's single, unique, absolute spacetime.*

This was the conclusion to which Einstein was driven in the summer of 1912 (though he preferred to use the word "curvature" rather than "warpage"). After four years of ridiculing Minkowski's idea of absolute spacetime, Einstein had finally been driven to embrace it, and warp it.

What does it mean for spacetime to be curved (or warped)? For clarity, ask first what it means for a two-dimensional surface to be curved (or warped). Figure 2.4 shows a flat surface and a curved surface. On the flat surface (an ordinary sheet of paper) are drawn two absolutely straight lines. The lines start out side by side and parallel. The ancient Greek mathematician Euclid, who created the subject now called "Euclidean geometry," used as one of his geometric postulates the demand that two such initially parallel lines never cross. This non-crossing is an unequivocal test for the flatness of the surface on which the lines are drawn. If space is flat, then initially parallel straight lines can never cross. If we ever find a pair of initially parallel straight lines that do cross, then we will know that space is not flat.

The curved surface in Figure 2.4 is a globe of the Earth. Locate on that globe the city of Quito, Equador; it sits on the equator. Send out a precisely straight line from Quito, directed northward. The line will travel northward, at constant longitude, through the North Pole.

In what sense is this a straight line? In two senses. One is the sense so crucial to airlines: It is a great circle, and the great circles on the Earth's globe are the shortest routes between two points and thus are the kinds of routes along which airlines like to fly. Construct any other line connecting Quito to the North Pole; it will necessarily be longer than the great circle.